

Democratizing Data: A deep dive into Eurostat Open Database



Who I am



Master's degree in Digital
Humanities - Language
Technologies at University of Pisa



Data Scientist at Clearbox AI



Outside of work, I'm an organizer
Python Torino community and a
volunteer for Pycon Italia.



Github Repo with slides and notebook





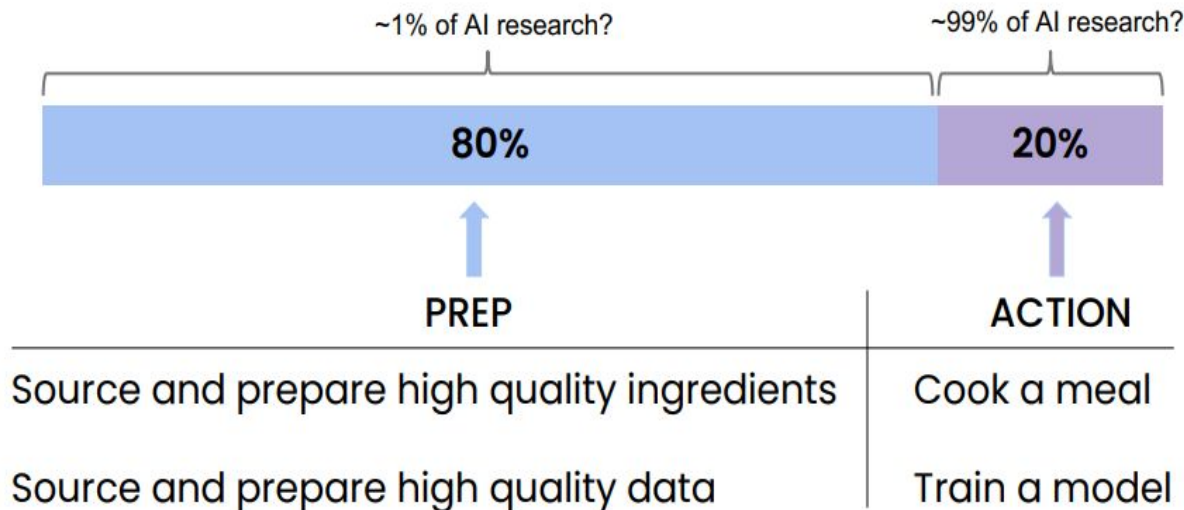
AI = Data + Code + Compute



AI needs Data

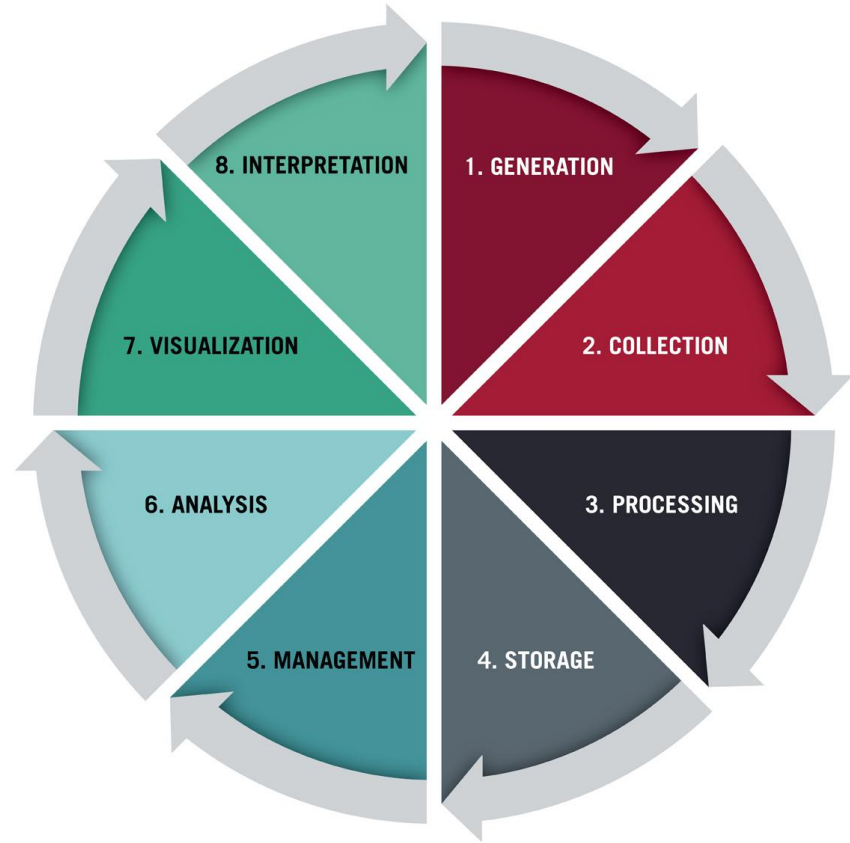


Data is Food for AI

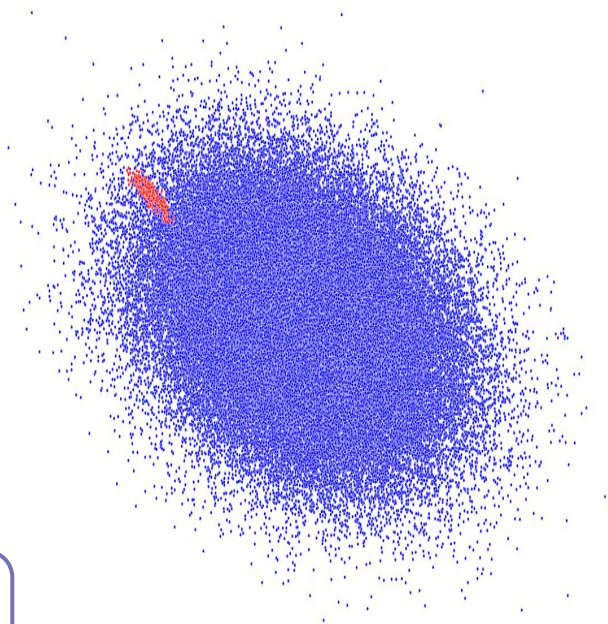


80% of AI work is actually **data work**

Biggest reason for AI project failure is data (quantity and quality)



Data Problems



SCARCITY

Data can be poor or strongly **imbalanced**

COSTS

Data collection can be expensive

ACCESSIBILITY

of data used to train AI models (Data protection **regulations** such as GDPR/HIPAA/AI-Act)



Public Available Databases



eurostat 

 DATA.GOV



International
Labour
Organization

ILOSTAT

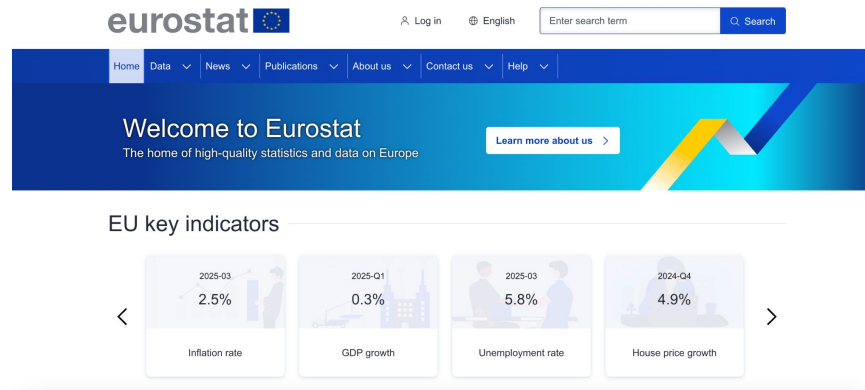


Istat

Istituto Nazionale
di Statistica

Eurostat

- [Eurostat](https://ec.europa.eu/eurostat) is the statistical office of the European Union.
- It offers a vast array of datasets covering everything from population dynamics and economic indicators to health, education, and environmental data.



Navigating Eurostat Database



<https://ec.europa.eu/eurostat/web/main/home>



Welcome to Eurostat

The home of high-quality statistics and data on Europe

[Learn more about us](#)

EU key indicators



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\(CROS\)](#)[Evaluations](#)

EU



2024-Q4

4.9%

House price growth



Database

 Information on the database

 Data navigation tree

 Detailed datasets

 General and regional statistics

 Economy and finance

 Population and social conditions

 Population and housing censuses (cens)

 Demography, population stock and balance (demo)

 Population projections (proj)

 Migration (migr)

 Health (hlth)

 Disability (dsb)

 Education and training (educ)

 Labour market (labour)

 Living conditions and welfare (livcon)

 Income, consumption and wealth - experimental statistics (icw) 

 Social protection (spr)  [\(Information note\)](#)



Data Browser

[Database](#) [Recently updated](#) [Bulk download](#) [Info](#) [Help](#)[Home](#) > [Database](#) > [Population and social conditions](#) > [Migration](#) > [International migration, citizenship](#) > [Immigration](#)

Immigration by age group, sex and citizenship

Online data code: [migr_imm1ctz](#) | DOI: [10.2908/migr_imm1ctz](#) | last update: 10/04/2025 23:00 | view: DEFAULT

Source of data: Eurostat

 Metadata Dataset information Add to 'My datasets' Selection[Customise your dataset](#)

Row

 Geopolitical entity (reporting) 

49/49 values displayed

Drag and drop here for breakdown

Column

 Time 

10/26 values displayed

Drag and drop here for breakdown

Dimensions

 Country of citizenship 

Total - 6/289 values displayed

Time frequency: [Annual](#)Unit of measure: [Number](#)Age definition: [Age reached during the year](#) Age class: [Total](#) Sex: [Total](#) 

Table

Line

Bar

Map

Display settings

Highlight

Quick selection

Default view

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Show legend

TIME	2018	2019	2020	2021	2022	2023
GEO						
European Union - 27 countries (from 2020)	(ep) 3 906 083 (ep)	4 176 776 (bep)	3 255 433 (ep)	4 054 686 (ep)	6 985 319 (ep)	5 934 699 (p)
Belgium	137 860	150 006	118 683	139 743	288 356	169 609
Bulgaria	(p) 29 559 (p)	37 929 (p)	37 364 (p)	39 461	40 619	56 807
Czechia	65 910	105 888	63 095	69 360	347 429	142 149
Denmark	64 669	61 384	57 230	63 489	121 183	98 344
Germany	(e) 825 736 (e)	804 821 (e)	642 399 (e)	775 946 (e)	1 943 445 (e)	1 271 219
Estonia	17 547	18 259	16 209	19 524	49 414	26 399
Ireland	105 699	96 321	81 918	83 770	157 537	135 617
Greece	119 489	129 459	84 221	57 120	96 662	118 816
Spain	643 684	750 480	467 918	887 960	1 258 894	1 250 991
France	387 160	385 596	331 137	336 401	431 078 (p)	417 613 (p)
Croatia	26 029	37 726	33 414	35 912	57 972	69 396
Italy	332 324	332 778	247 526	318 366	410 985	439 658
Cyprus	25 740	27 553	22 850	25 473	37 558	40 761
Latvia	10 909	11 223	8 840	12 689	38 708	18 710
Lithuania	37 420	46 526	46 020	44 858	87 367	66 682
Luxembourg	24 644	26 668	22 490	25 335	31 433	26 964
Hungary	82 937	88 581	75 470	80 471	94 148	96 192
Malta	27 394	29 384	14 751	16 963	34 964	42 239
Netherlands	194 306	215 756	182 244	214 105	326 798	274 597
Austria	105 633	109 167	103 565	118 511	201 622	147 591
Poland	(ep) 214 083 (ep)	226 649 (ep)	210 615 (ep)	241 116	275 515 (p)	303 275 (p)
Portugal	55 357	95 382	83 654	97 119	167 098	189 367
Romania	(e) 172 578 (e)	202 422 (e)	145 519 (e)	194 642	293 024 (e)	324 091
Slovenia	28 455	31 319	36 110	23 624	35 613	33 939
	(n) 7 253 (n)	7 016 (n)	6 775 (n)	5 733	5 463	5 923



Using Python with Eurostat



Eurostat Python Package

- Developed by the European Commission Joint Research Center



eurostat 1.1.1

```
pip install eurostat
```



[Python Package](#)
[Colab Notebook](#)

Takeaways



- **Data** is the foundation of every AI system. Without high-quality, accessible data, even the best models can't deliver meaningful results.
- **Publicly available datasets** like Eurostat offer a **transparent**, **reliable**, and **ethical** alternative to private data sources.
- Knowing how to navigate tools like the Eurostat interface and the Eurostat Python Package is important to get **meaningful insights**.
- Open data empowers a broader community to **contribute to responsible and inclusive AI development**.



Any questions?





Thanks for Reading

Feel free to contact us:



www.clearbox.ai



simona@clearbox.ai
info@clearbox.ai